

Probability and Probability Distributions

6 (2)

Complement

Complement of an event

Definition 8:- The set of all sample events in the sample space that do not belong to event A is called the complement of event A and is denoted by A' or $\bar{A}$ or sometimes A^c .

If two events A and A' are complementary, then

$$P(A \cup A') = P(A) + P(A') = P(S) = 1$$

$$i.e. P(A) = 1 - P(A').$$

Complement of an event

Example 9:- Toss a coin once and find the probability that the result on top is not a head.

Example 10:- Draw a card once from a well shuffled ordinary deck of 52 cards. Find the probability that the drawn card is

- | | |
|----------------------|----------------------|
| (i) not a black card | (iii) not a diamond |
| (ii) not a King | (iv) not a face card |

*Dependent events,
Independent events and
Conditional Probability*

6.2 *Dependent events, Independent events and Conditional Probability*

Definition 9:- Two events are said to be independent, if the occurrence or non-occurrence of one event has no effect on the probability of the other event. Otherwise they are called dependent events.

For events to be considered dependent, one must have an influence over how probable another is.

6.2 Dependent events, Independent events and Conditional Probability

For example, say you'd like to go on vacation to Bahir Dar at the end of this term , it depends on the security on the way and around the city of Bahir Dar, having enough money . . .

An event is deemed *independent* when it is not connected to another event, or its probability of happening, or conversely, of not happening.

6.2 Dependent events, Independent events and Conditional Probability

This is true of events in terms of probability, as well as in real life, which, as mentioned above, is true of dependent events as well.

For example, the color of your hair has absolutely no effect on where you learn. The two events of “*having black hair*” and “*learning at HiLCoE*” are completely independent of one another.

6.2 Dependent events, Independent events and Conditional Probability

Example 11:- Toss a coin twice. Define events

A and B as:

A: The first toss results a head

B: The second toss results a head

Here we can see that the result of the first toss (what so ever it is) does not affect the probability of occurrence of a head on the second toss.

6.2 Dependent events, Independent events and Conditional Probability (Contd. ...)

That is whether the first toss is a head or not, the probability of getting a head on the second toss is still half. Thus A and B are independent events.

Example 12:- Draw two cards turn by turn without replacement from a well shuffled ordinary deck of 52 cards. Define events C and D as:

6.2 Dependent events, Independent events and Conditional Probability (Contd. ...)

C: the first card is a King

D: the second card is a King

In this case we can see that $P(C) = \frac{n(C)}{n(S)} = \frac{4}{52}$.

Now suppose that C occurred. That is the first card is a King. Then we are left with 51 cards out of which 3 are Kings.

6.2 Dependent events, Independent events and Conditional Probability (Contd. ...)

So the probability that the second card is a king will be $\frac{3}{51}$, i.e. $P(D) = \frac{3}{51}$.

$\therefore C$ and D are dependent events

6.2 Dependent events, Independent events and Conditional Probability (Contd. ...)

However, if C didn't occur, i.e. if the first drawn card was not a king, then we are left with 51 cards out of which 4 are kings. So the probability that the second card is a king will be $\frac{4}{51}$.

Hence the probability of the second card to be a King is not fixed but depends on the occurrence or non- occurrence of event C.

6.2 Dependent events, Independent events and Conditional Probability (Contd. ...)

Definition 10:- When two events are dependent, the probability of one event given that the other event has occurred is called **conditional probability.**

Notation:- $P(B \setminus A)$ denotes the probability of event B given that A has occurred.

6.2 Dependent events, Independent events and Conditional Probability (Contd. ...)

□ If two events A and B are independent, then the probability of getting both A and B is defined by:

$$*P(A \text{ and } B) = P(A) \times P(B).*$$

□ If two events A and B are dependent, then the probability of getting both A and B is defined by:

$$*P(A \text{ and } B) = P(A) \times P(B \setminus A).*$$

6.2 Dependent events, Independent events and Conditional Probability (Contd. ...)

Example 13:- Flip a coin three times. What is the probability of getting a tail in all the three tosses?

Example 14:- Draw two cards turn by turn without replacement from a well shuffled ordinary deck of 52 cards. What is the probability of getting

- (i) an Ace in both draws?
- (ii) a Face card in both draws?

6.2 Dependent events, Independent events and Conditional Probability

Example 14:- A bag contains 4 red, 2 green 7 blue balls. What is the probability of drawing red ball in the second draw given that the first ball drawn is blue?

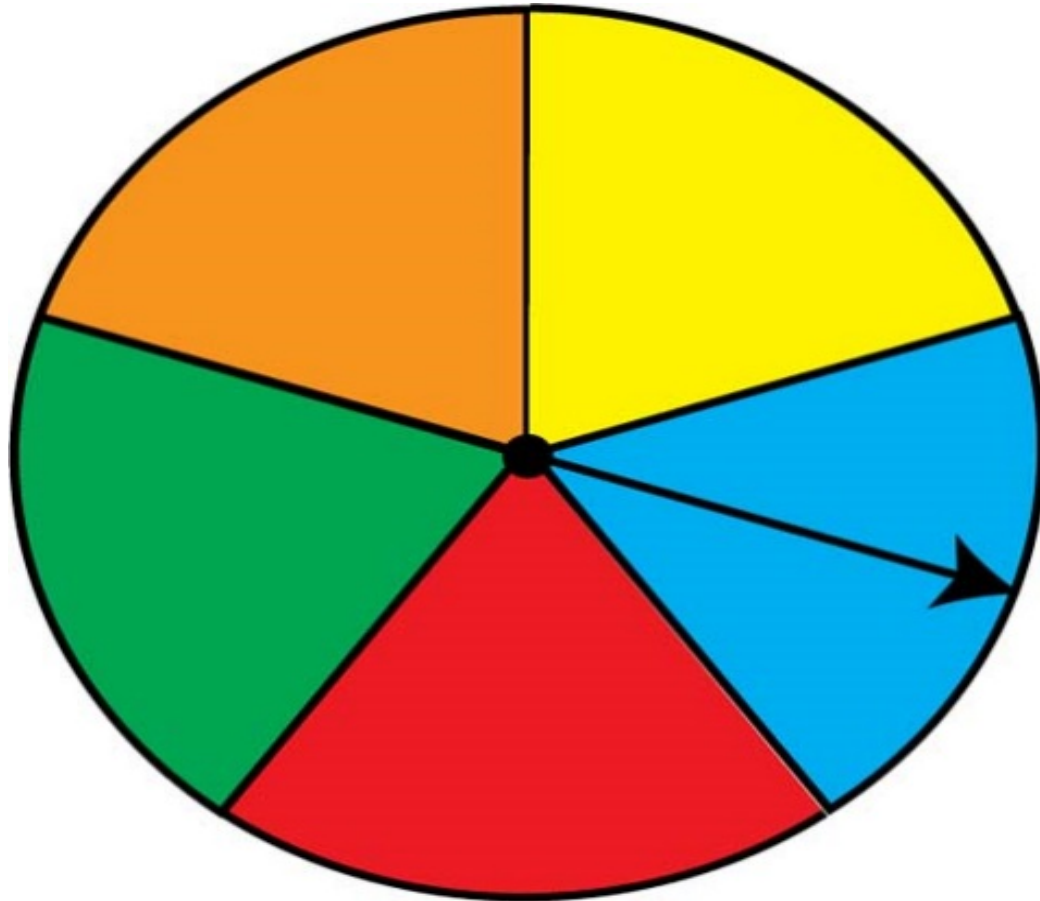
- i) The balls drawn are replaced in the bag.
- ii) The balls are drawn turn by turn without replacement.

Complement of an event

- Example 11:--** A bag contains 5 green buttons, 2 blue buttons and 6 white buttons. Maya takes two buttons from the bag at random turn by turn without replacement. Calculate the probability that
- the first button is green and the second is not green.
 - one button is green and the other button is not green.

Complement of an event

Example 12:- A spinner has five sides. Each side is painted red, blue, green, yellow or orange.



Complement of an event

The table shows some of the probabilities of the spinner on each colour.

Colour	Red	Blue	Green	Yellow	Orange
Probability	0.3	0.16	0.18	0.25	

- a) Complete the table.
- b) Meron spins the spinner once. Find the probability that the spinner lands on red or blue.

Complement of an event

Example 13:- Bag A and bag B each contain red sweets and yellow sweets. Radia picks a sweet at random from bag A. Solomon picks a sweet at random from bag B.

The probability that Radia picks a red sweet is $\frac{2}{5}$.

The probability that Radia and Solomon both pick a

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Complement of an event

Solution 12:- Given that P

$$P(R_A) = \frac{2}{5}$$

$$\Rightarrow P(Y_A) = 1 - P$$

Any Questions



Any Questions

